SwarmSim: Test Report

Simulation-based assessment of two multi-UAV formation controllers under injected faults. Simulation only: no flight hardware, payload or operational capability.

1. Summary

- 23 of 24 scenario/controller combinations met the acceptance rule (trial pass rate at or above the required level).
- Leader-follower passed 343 of 360 trials overall (95.3%, Wilson 95% interval 92.6% to 97.0%).
- Consensus passed 357 of 360 trials overall (99.2%, Wilson 95% interval 97.6% to 99.7%).
- NOT ACCEPTED: 11 high wind with Leader-follower: 19/30 passed; failed criteria: steady_error_p95_m (11x).
- Individual failed trials inside accepted combinations: geofence breach: 04 gps dropout leader/Consensus x1, 04 gps dropout leader/Leader-follower x1; min separation m: 12 compound/Leader-follower x1; recovery time s: 05 comms loss single/Consensus x1, 08 drone failure leader/Leader-follower x1; steady error p95 m: 02 gps dropout short/Leader-follower x2, 08 drone failure leader/Leader-follower x2, 11 high wind/Consensus x1.
- Collisions (two drones closer than 2 m): 1 of 720 trials with the safety layer on.
- Controller differences larger than the statistical noise (non-overlapping 95% intervals): 11 high wind (Consensus better: 19/30 vs 29/30).
- Safety-layer ablation (same seeds, 480 trials each way): 469 passed with the layer on and 469 with it off; collisions 0 on vs 1 off; trials inside the 4 m margin 4 on vs 10 off.
- Scenarios where switching the layer off made the worst case clearly worse (more than 0.5 m): 05 comms loss single (worst separation 6.0 m with, 1.0 m without); 06 comms blackout all (worst separation 5.1 m with, 4.6 m without); 10 packet loss 30 (worst separation 4.8 m with, 3.9 m without); 12 compound (worst separation 4.0 m with, 3.2 m without). Elsewhere the layer changed the worst case by less than 0.5 m.
- False peer-loss alarms (heartbeat timeout with no fault involved): 0 of 720 trials.

2. Method

- 12 scenarios x 2 controllers x 30 seeded trials = 720 trials. Seeds are fixed, so every number is reproducible.
- Each trial flies a 5-drone wedge on a square mission (about 100 s) with wind, GPS noise, a lossy radio and a fault schedule. Fault onset times carry a seeded jitter.
- Acceptance criteria were fixed in criteria/criteria.yaml before the campaign. A trial passes when every applicable criterion passes; a combination is accepted when at least 90% of trials pass.
- Metrics are computed from ground truth, never from what a drone believes. On-board code only uses its own estimates and received messages.

3. Results

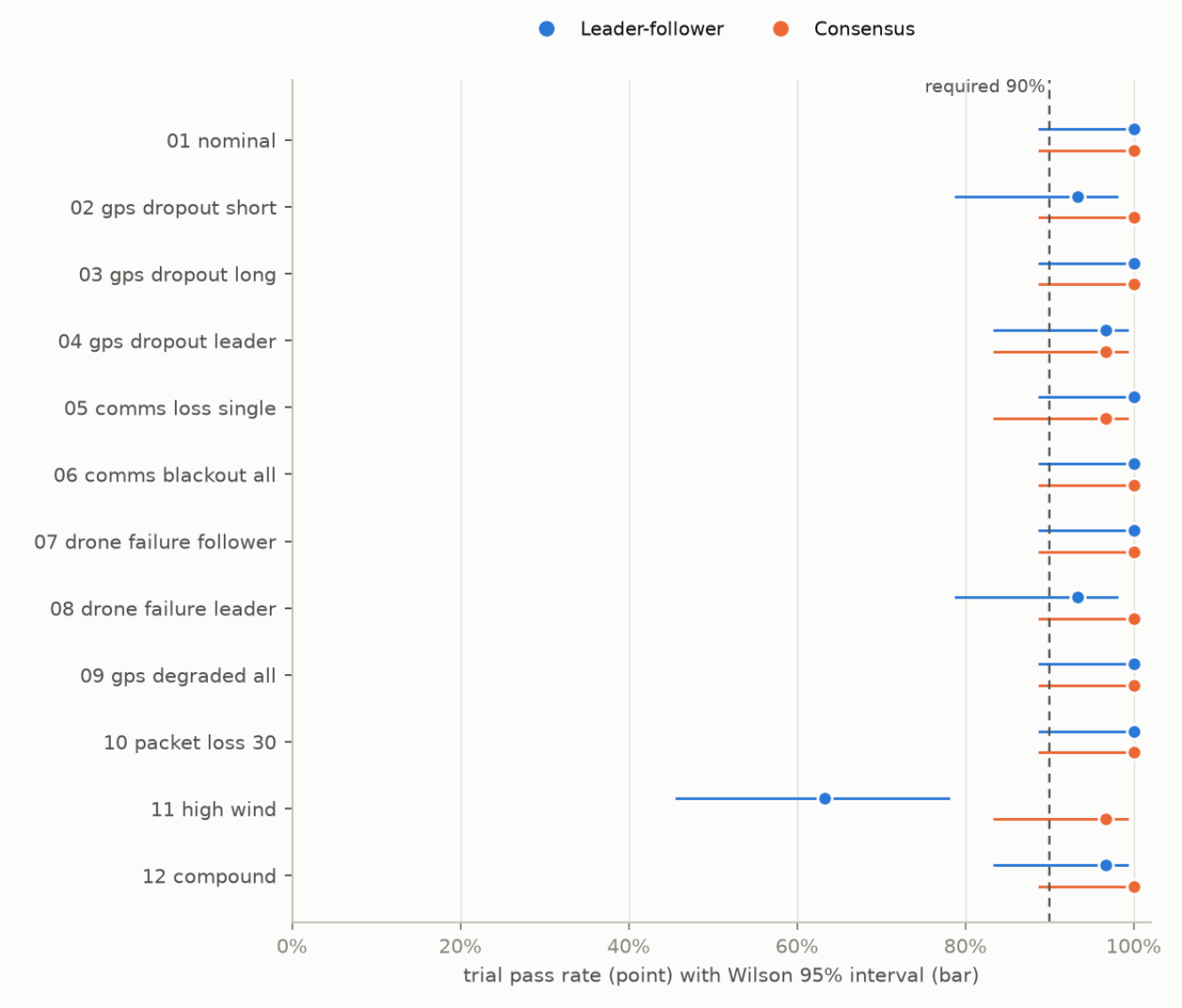


Figure 1. Trial pass rate with Wilson 95% interval. The dashed line is the required rate.

Scenario	Controller	Pass	Rate [95% CI]	Verdict	Min sep [m]	Coll.	Detect p95 [s]	Recover p95 [s]	Steady err p95 [m]
01 nominal	Consensus	30/30	100% [89%-100%]	PASS	7.60	0	-	-	1.63
01 nominal	Leader-follower	30/30	100% [89%-100%]	PASS	6.27	0	-	-	1.96
02 gps dropout short	Consensus	30/30	100% [89%-100%]	PASS	3.69	0	2.0	2.1	1.62
02 gps dropout short	Leader-follower	28/30	93% [79%-98%]	PASS	3.93	0	2.0	2.1	2.86
03 gps dropout long	Consensus	30/30	100% [89%-100%]	PASS	3.69	0	2.0	2.1	1.67
03 gps dropout long	Leader-follower	30/30	100% [89%-100%]	PASS	3.93	0	2.0	2.1	2.05
04 gps dropout leader	Consensus	29/30	97% [83%-99%]	PASS	3.83	0	1.9	2.1	1.73
04 gps dropout leader	Leader-follower	29/30	97% [83%-99%]	PASS	4.41	0	1.9	2.1	2.37
05 comms loss single	Consensus	29/30	97% [83%-99%]	PASS	7.24	0	2.8	3.1	1.68
05 comms loss single	Leader-follower	30/30	100% [89%-100%]	PASS	6.03	0	2.8	3.1	1.94
06 comms blackout all	Consensus	30/30	100% [89%-100%]	PASS	7.54	0	2.4	8.4	1.58
06 comms blackout all	Leader-follower	30/30	100% [89%-100%]	PASS	5.09	0	2.4	8.8	2.24
07 drone failure follower	Consensus	30/30	100% [89%-100%]	PASS	7.75	0	2.8	2.8	1.60
07 drone failure follower	Leader-follower	30/30	100% [89%-100%]	PASS	6.26	0	2.8	2.8	1.87
08 drone failure leader	Consensus	30/30	100% [89%-100%]	PASS	5.58	0	2.8	9.1	2.09
08 drone failure leader	Leader-follower	28/30	93% [79%-98%]	PASS	5.79	0	2.8	11.4	3.42

Scenario	Controller	Pass	Rate [95% CI]	Verdict	Min sep [m]	Coll.	Detect p95 [s]	Recover p95 [s]	Steady err p95 [m]
09 gps degraded all	Consensus	30/30	100% [89%-100%]	PASS	5.82	0	-	-	1.88
09 gps degraded all	Leader-follower	30/30	100% [89%-100%]	PASS	4.07	0	-	-	2.51
10 packet loss 30	Consensus	30/30	100% [89%-100%]	PASS	7.90	0	-	-	1.64
10 packet loss 30	Leader-follower	30/30	100% [89%-100%]	PASS	4.76	0	-	-	2.34
11 high wind	Consensus	29/30	97% [83%-99%]	PASS	7.55	0	-	-	3.62
11 high wind	Leader-follower	19/30	63% [46%-78%]	FAIL	6.46	0	-	-	8.19
12 compound	Consensus	30/30	100% [89%-100%]	PASS	4.75	0	2.7	2.8	1.73
12 compound	Leader-follower	29/30	97% [83%-99%]	PASS	1.24	1	2.7	2.8	2.58

Table 1. Per scenario and controller. Min sep is the smallest distance between any two drones in any trial. Detect, recover and steady error are 95th percentiles over trials; '-' means not applicable, 'none' means it never happened in some trial.

4. Safety

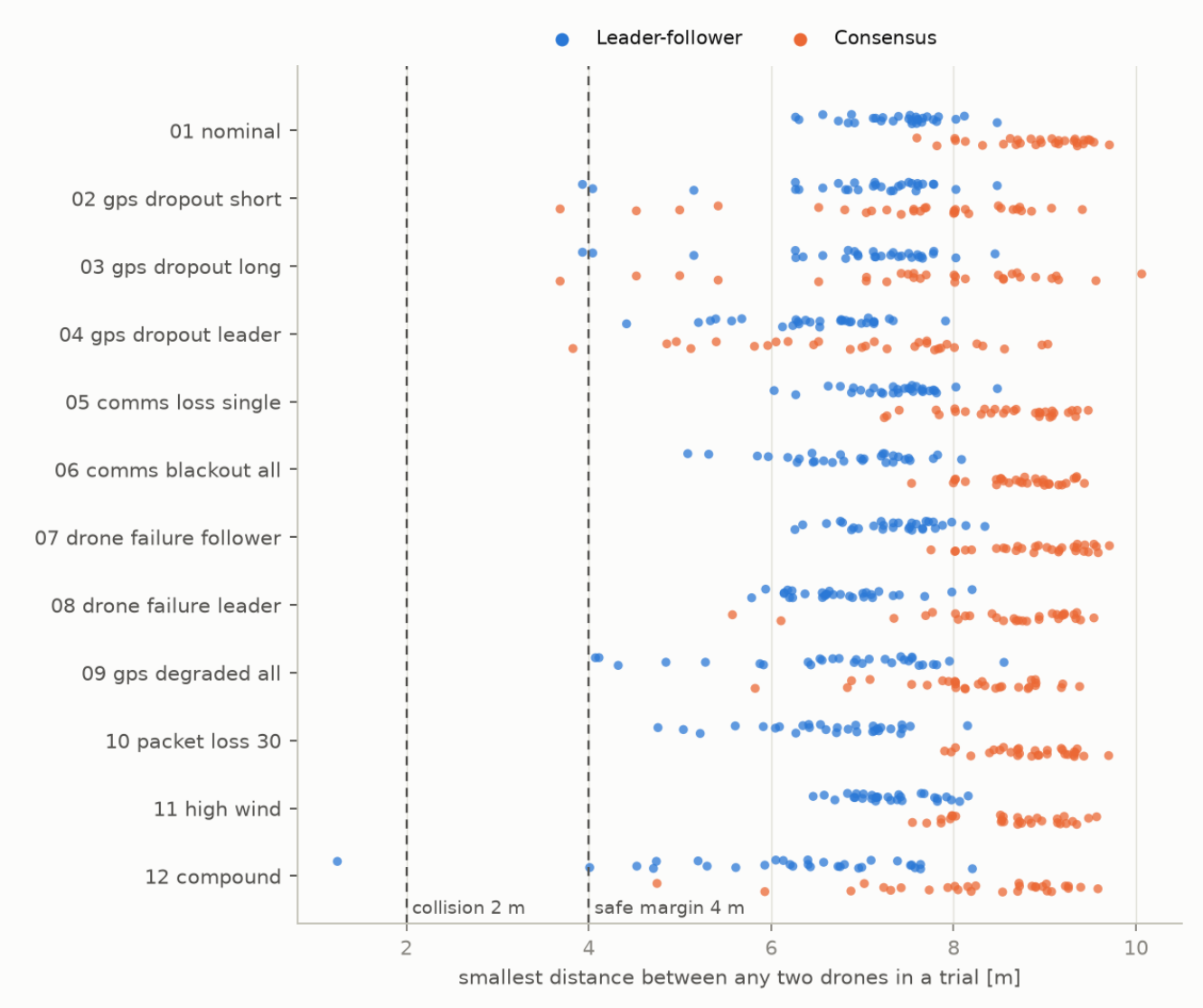


Figure 2. Smallest distance between any two drones in each trial (one dot per trial). Left of the 2 m line is a collision.

Switching the runtime safety layer (separation keeping, tracks of silent peers, vertical deconfliction) off on the same seeds shows where it changes the outcome and where it does not.

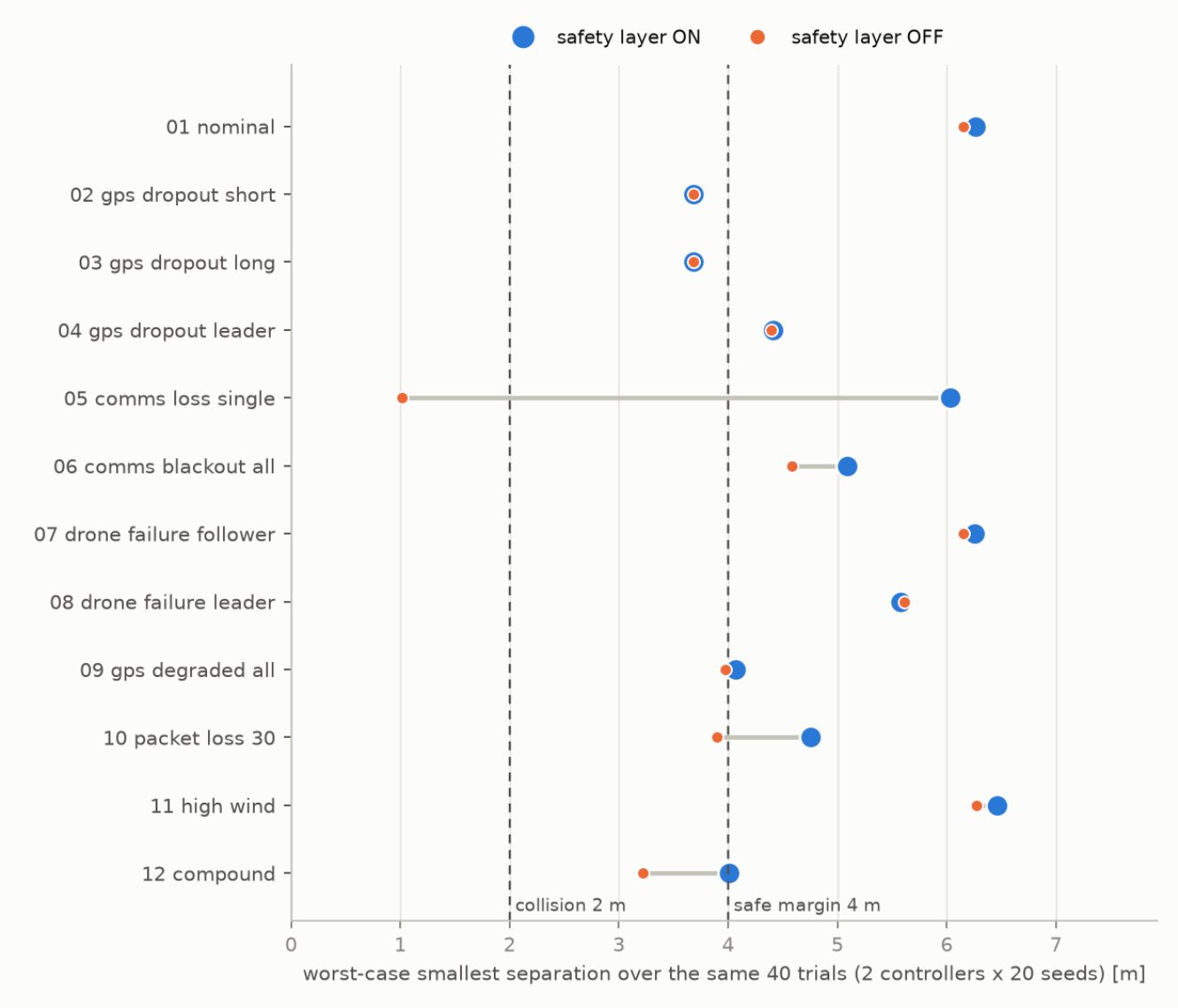


Figure 3. Worst-case smallest separation, safety layer on versus off, same seeds. Dots on top of each other mean no difference.

5. Fault handling and accuracy

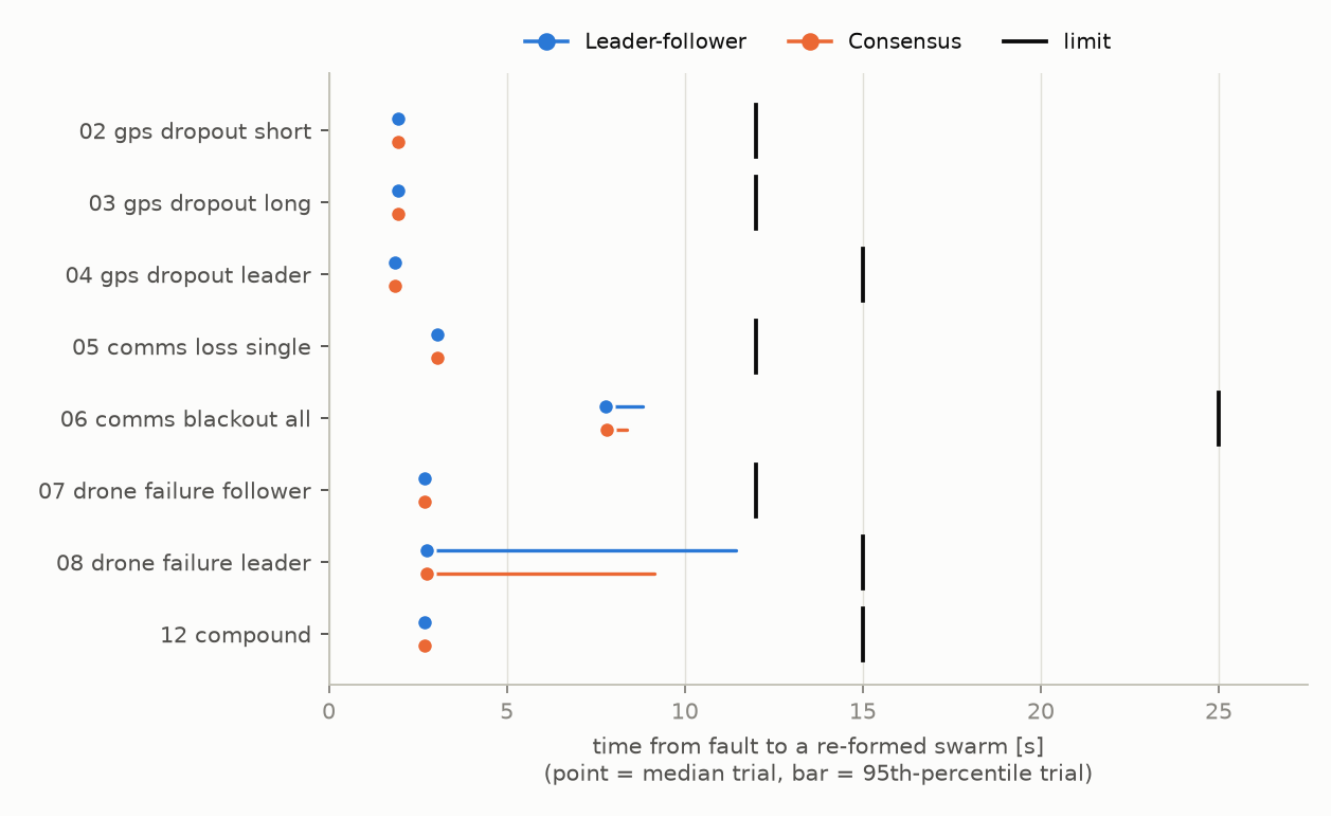


Figure 4. Time from fault onset until the surviving swarm is formed up again and everyone agrees who is in it. Black ticks are the scenario limits.

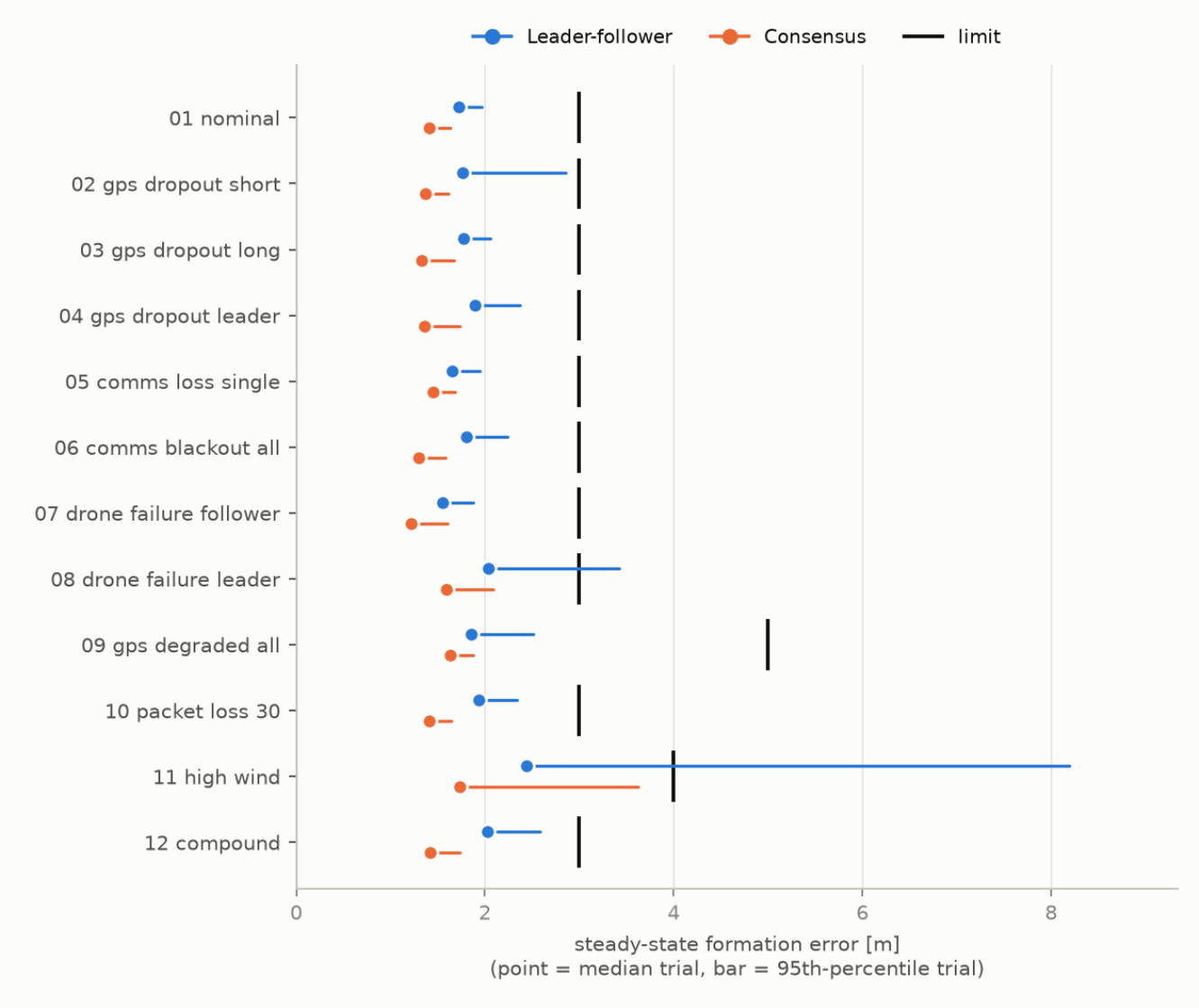


Figure 5. Formation shape error in steady state. Black ticks are the scenario limits.

6. Example trial

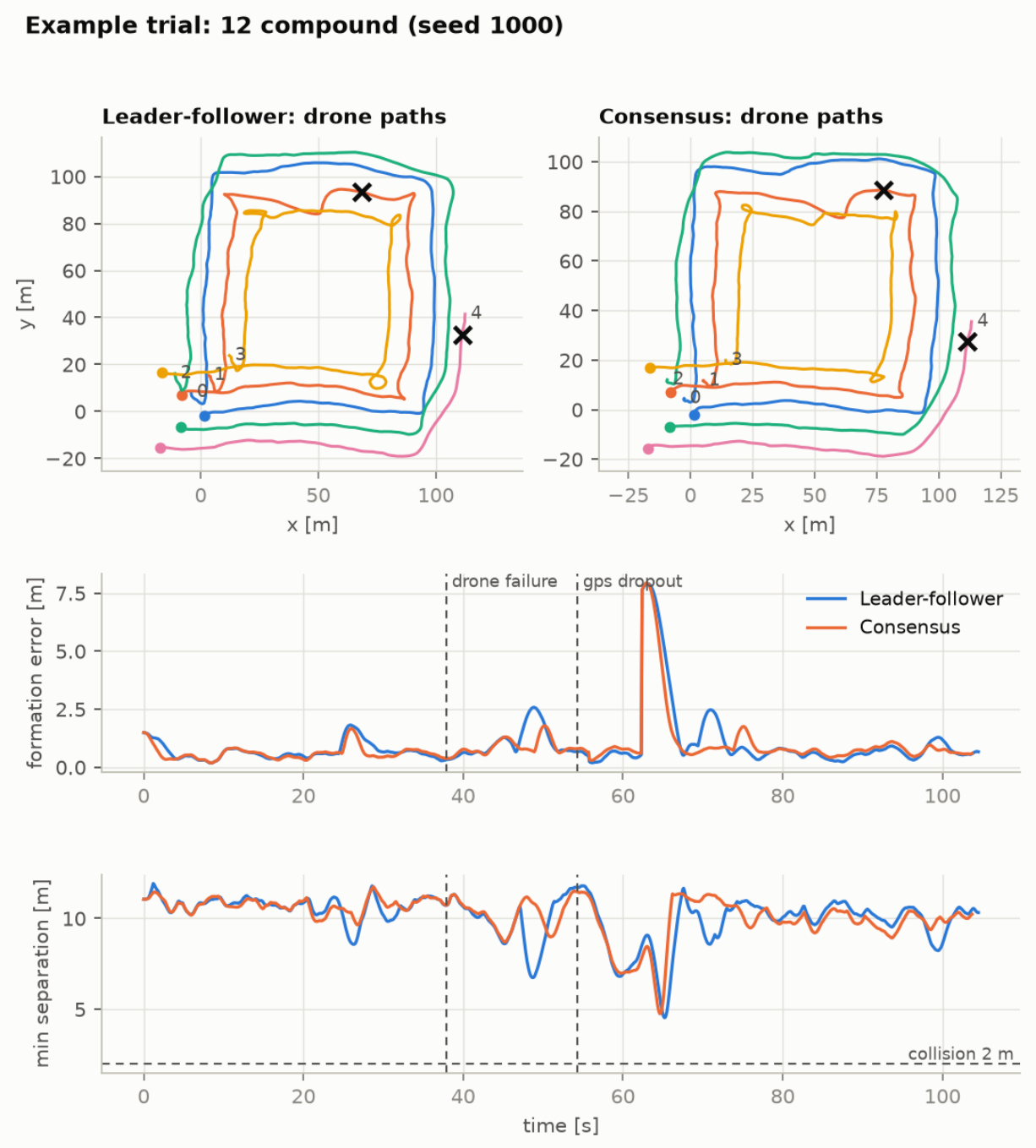


Figure 6. The compound scenario (packet loss, a follower crash, then a GPS dropout) for both controllers. Crosses mark where drones were hit by a fault.

7. Acceptance criteria applied

Scenario	Criteria
01 nominal	min_separation_m >= 2.0; time_below_safe_sep_s <= 1.0; steady_error_p95_m <= 3.0; formation_error_max_m <= 25.0; mission_complete = True; geofence_breach = False; false_peer_loss_events <= 0; survivors_at_end >= 5
02 gps dropout short	min_separation_m >= 2.0; time_below_safe_sep_s <= 1.0; steady_error_p95_m <= 3.0; formation_error_max_m <= 25.0; mission_complete = True; geofence_breach = False; detection_time_s <= 4.0; recovery_time_s <= 12.0; rejoin_time_s <= 30.0; survivors_at_end >= 5
03 gps dropout long	min_separation_m >= 2.0; time_below_safe_sep_s <= 1.0; steady_error_p95_m <= 3.0; formation_error_max_m <= 25.0; mission_complete = True; geofence_breach = False; detection_time_s <= 4.0; recovery_time_s <= 12.0; survivors_at_end >= 4
04 gps dropout leader	min_separation_m >= 2.0; time_below_safe_sep_s <= 1.0; steady_error_p95_m <= 3.0; formation_error_max_m <= 25.0; mission_complete = True; geofence_breach = False; detection_time_s <= 4.0; recovery_time_s <= 15.0; survivors_at_end >= 4

Scenario	Criteria
05 comms loss single	min_separation_m >= 2.0; time_below_safe_sep_s <= 1.0; steady_error_p95_m <= 3.0; formation_error_max_m <= 25.0; mission_complete = True; geofence_breach = False; detection_time_s <= 4.0; recovery_time_s <= 12.0; rejoin_time_s <= 30.0; survivors_at_end >= 5
06 comms blackout all	min_separation_m >= 2.0; time_below_safe_sep_s <= 1.0; steady_error_p95_m <= 3.0; formation_error_max_m <= 25.0; mission_complete = True; geofence_breach = False; detection_time_s <= 4.0; recovery_time_s <= 25.0; rejoin_time_s <= 30.0; survivors_at_end >= 5
07 drone failure follower	min_separation_m >= 2.0; time_below_safe_sep_s <= 1.0; steady_error_p95_m <= 3.0; formation_error_max_m <= 25.0; mission_complete = True; geofence_breach = False; detection_time_s <= 4.0; recovery_time_s <= 12.0; survivors_at_end >= 4
08 drone failure leader	min_separation_m >= 2.0; time_below_safe_sep_s <= 1.0; steady_error_p95_m <= 3.0; formation_error_max_m <= 25.0; mission_complete = True; geofence_breach = False; detection_time_s <= 4.0; recovery_time_s <= 15.0; survivors_at_end >= 4
09 gps degraded all	min_separation_m >= 2.0; time_below_safe_sep_s <= 1.0; steady_error_p95_m <= 5.0; formation_error_max_m <= 30.0; mission_complete = True; geofence_breach = False; false_peer_loss_events <= 0; survivors_at_end >= 5
10 packet loss 30	min_separation_m >= 2.0; time_below_safe_sep_s <= 1.0; steady_error_p95_m <= 3.0; formation_error_max_m <= 25.0; mission_complete = True; geofence_breach = False; false_peer_loss_events <= 0; survivors_at_end >= 5
11 high wind	min_separation_m >= 2.0; time_below_safe_sep_s <= 1.0; steady_error_p95_m <= 4.0; formation_error_max_m <= 25.0; mission_complete = True; geofence_breach = False; false_peer_loss_events <= 0; survivors_at_end >= 5
12 compound	min_separation_m >= 2.0; time_below_safe_sep_s <= 1.0; steady_error_p95_m <= 3.0; formation_error_max_m <= 25.0; mission_complete = True; geofence_breach = False; detection_time_s <= 4.0; recovery_time_s <= 15.0; rejoin_time_s <= 30.0; survivors_at_end >= 4

8. Diagnostics after the campaign

Hand-written analysis of the campaign of 22 Oct 2026 (seeds 1000 to 1029). Not regenerated when the campaign is rerun.

- Strong wind, leader-follower (19 of 30 passed): the cause is airspeed headroom. A 3 to 5 m/s wind against a 4 m/s cruise speed needs about 7 to 9 m/s of air speed on upwind legs, at or above the 8 m/s vehicle limit. Rerunning the same 30 seeds with the limit raised to 12 m/s gives 30 of 30 for both controllers. Not fixed: slowing the mission to keep headroom would trade a formation failure for a mission-time failure.
- Geofence breach in the leader GPS-loss scenario (1 of 30 trials per controller): the drone that lost GPS drifted about 52 m on dead reckoning while landing and was outside the fence for 3 to 4 s at 3 to 4 m altitude. Not fixed.
- The single collision in 720 trials (compound scenario, leader-follower, 1.24 m): two healthy drones in formation, about 10 s after a GPS dropout ended. Cause not isolated.
- Safety layer: it changes the worst case clearly only where a neighbour goes silent (worst separation 6.0 m with it, 1.0 m without, in the single radio-loss scenario). Elsewhere total pass counts are identical with it on and off.

9. Limitations

- Simulation only. The vehicle is a point mass with a first-order velocity loop, an acceleration limit and a speed limit. It is not a rotor-level or attitude model and no real flight stack (PX4/ArduPilot) is in the loop.
- GPS, the radio and the wind are statistical models with the parameters listed in config.py. Real multipath, jamming, antenna patterns and terrain effects are not modelled.
- Results describe behaviour of this simulator under these faults. They are evidence for design decisions and a regression baseline, not a safety case or a guarantee about any physical system.
- Pass rates come from 30 trials per case, so a 100% rate still has a Wilson lower bound near 89%. The report therefore shows intervals, not only point values.
- Formation slots are fixed per drone: after a loss the formation keeps a gap instead of closing ranks. This was a deliberate choice to avoid slot swaps when a drone rejoins.
- Faults are injected one scenario at a time (plus one compound case). Common-cause failures and adversarial network behaviour are out of scope.
- The acceptance criteria were written before the campaign; two design parameters (heartbeat timeout, altitude layer in GPS failsafe) were changed after earlier small test runs exposed problems. The change history is in docs/DESIGN.md.

10. Reproducing this report

```
pip install -r requirements.txt
python -m swarmsim run --trials 30 --ablate --ablate-trials 20 --out results
python -m swarmsim report
python -m pytest tests -q
```

- Python 3.14.3, numpy 2.5.4, matplotlib 3.11.2
- Platform: Windows AMD64
- Generated: 2026-10-23 07:12